

SENTIMENT ANALYSIS ON PRODUCT REVIEWS

Week 1 Internship Project

OBJECTIVE

The objective of this project was to analyze customer reviews and classify them into Positive, Negative, and Neutral categories using Natural Language Processing (NLP). The goal was to understand customer opinions, measure customer satisfaction levels, and generate meaningful business insights from product reviews.

METHODOLOGY

The Amazon Fine Food Reviews dataset was used for this analysis. The first 5,000 reviews were selected and processed. Data cleaning was performed by removing missing and duplicate reviews. Sentiment analysis was carried out using the TextBlob library, which calculates sentiment polarity scores. Based on these scores, reviews were classified as Positive, Negative, or Neutral. Various visualizations were created to better understand sentiment distribution and customer feedback patterns.

FINDINGS

The analysis revealed that 88.34% of the reviews were positive, indicating a very high level of customer satisfaction. Most customers appreciated the product quality, taste, freshness, and overall value. Negative reviews mainly focused on packaging issues, delivery delays, product quality inconsistencies, and personal taste preferences. One surprising observation was that some low-rated reviews still contained positive language, demonstrating that customer sentiment can be more complex than ratings alone.

RECOMMENDATION

Based on the findings, the business should continue maintaining its high product quality standards while focusing on reducing packaging and delivery-related issues. Addressing

recurring customer complaints and improving the overall customer experience can further increase customer satisfaction, strengthen customer loyalty, and enhance brand reputation.

CONCLUSION

This project successfully demonstrated the use of sentiment analysis for understanding customer opinions at scale. By leveraging NLP techniques and data visualization, businesses can quickly identify customer satisfaction trends, monitor feedback, and make informed decisions to improve products and services. The results indicate that the majority of customers have a positive perception of the products, highlighting strong overall performance.